

AAE 507: Principles of Dynamics

Problem Set 5 & 6

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Out: ~~Sep 25~~ Sep 26, 2026

PS5 Due: ~~Oct 2~~ Oct 3, 2026 at 11:59 PM ET

PS6 Due: ~~Oct 9~~ Oct 10, 2026 at 11:59 PM ET
(submission at Brightspace>Content>Gradescope)

Use 1×10^{-12} for the integration tolerance of ordinary differential equations unless otherwise requested. When plotting trajectories in a 2-D or 3-D space, use equal axes for the x -, y -, and z -axes.

Problem Set 5

(5.1) Orbital Dynamics with J_2 and Atmospheric Drag

(a): **[AI use Level: 0] Cartesian Lagrange model.**

Consider a spacecraft of mass m orbiting Earth perturbed by J_2 and atmosphere drag. Let the inertial frame be $\mathcal{N} : \{\hat{e}_x, \hat{e}_y, \hat{e}_z\}$ and write its Cartesian position as $\mathbf{r} = x\hat{e}_x + y\hat{e}_y + z\hat{e}_z$, $r = \|\mathbf{r}\|_2$. Expressing the potential energy for Earth gravity under J_2 in the Cartesian coordinates, we have

$$V(x, y, z) = -\frac{\mu m}{r} + \frac{\mu m J_2 r_o^2}{2r^5} (3z^2 - r^2), \quad (1)$$

where μ is Earth's gravitational parameter and $r_o = R_{\text{earth}}$ is its equatorial radius.

We use the same model as considered in PS1 to model the drag, which assumes an inertially stationary, spherically symmetric atmosphere, given by

$$\mathbf{F}_D = m\mathbf{a}_D, \quad \mathbf{a}_D = -\frac{1}{2}C_D \frac{A}{m} \rho(r) v^2 \hat{\mathbf{v}}, \quad v := \|\mathbf{v}\|_2, \quad \hat{\mathbf{v}} := \frac{\mathbf{v}}{v}, \quad (2)$$

where $\mathbf{v} = \dot{\mathbf{r}}$, C_D is the drag coefficient, A is the reference area, and

$$\rho(r) \approx \rho_0 \exp\left[-\frac{r - 6978 \text{ km}}{H}\right]. \quad (3)$$

(a.1): **Form** the Lagrangian using the Cartesian coordinates as generalized coordinates, $\mathbf{q} = [x, y, z]^\top$.

Hint: atmospheric drag, Eq. (2), is a non-conservative force.

(a.2): **Derive** the generalized force for Eq. (2) associated with $\mathbf{q} = [x, y, z]^\top$.

(a.3): **Derive** the Lagrange's equations and write the result as the Cartesian vector EOM

$$\ddot{\mathbf{r}} = \mathbf{a}_{2B} + \mathbf{a}_{J_2} + \mathbf{a}_D. \quad (4)$$

(b): **[AI use Level: 0] Spherical-coordinate Lagrange model.**

Now consider spherical coordinates $\mathbf{q} = [r, \phi, \theta]^\top$, where r is radius, ϕ is azimuth about $\hat{e}_z$, and θ is elevation measured from $\hat{e}_x$ - $\hat{e}_y$ plane. The potential energy for gravity with J_2 is given in class.

(b.1): **Form** the Lagrangian using the spherical coordinates.

- (b.2): **Derive** the generalized force for Eq. (2) associated with $\mathbf{q} = [r, \phi, \theta]^\top$.
- (b.3): **Derive** the Lagrange's equations and write the result as $\ddot{r} = ?$, $\ddot{\phi} = ?$, and $\ddot{\theta} = ?$.
- (b.4): For zero drag, **identify** the cyclic coordinate and the associated conserved generalized momentum. **Discuss** its relation to the angular momentum about $\hat{e}_z$, denoted by H_z .
- (b.5): For zero drag, **show** that the Jacobi energy integral (note different notation from class)

$$\mathcal{J} := \dot{\mathbf{q}}^\top \frac{\partial L}{\partial \dot{\mathbf{q}}} - L \quad (5)$$

is identical to the total energy $E := T + V$. **Discuss** if $\mathcal{J}$ is conserved under no drag, and why.

- (b.6): Under drag, **derive** the time derivatives of H_z and $\mathcal{J}$. **Discuss** if they are conserved, and why.
- (c): **[AI use Level: 1] Numerical propagation and verification.**

Let us now numerically simulate the orbital dynamics. For the physical parameters and initial condition, use the same values as defined in PS1, except for the initial semi-major axis, for which use $a = 6628$ km (instead of 6978 km).

- (c.1): **Express** the initial condition in the Cartesian state $(x, y, z, \dot{x}, \dot{y}, \dot{z})$ and spherical state $(r, \phi, \theta, \dot{r}, \dot{\phi}, \dot{\theta})$.
- (c.2): **Propagate** the orbit for 5 days using the EOM derived for the spherical coordinates, with the following two models: (i) J_2 gravity only, and (ii) J_2 gravity with drag. **Plot** each inertial trajectory in Cartesian coordinates separately. Also, **plot** the time history of the difference between the two models.
- (c.3): For both models, **plot** the normalized changes in $H_z, \mathcal{J}$:

$$\frac{\mathcal{J}(t) - \mathcal{J}(0)}{|\mathcal{J}(0)|}, \quad \frac{H_z(t) - H_z(0)}{|H_z(0)|}. \quad (6)$$

Discuss why the J_2 -only result makes sense using the symmetry of the potential. For the drag result, **compare** the signs and qualitative behavior against dH_z/dt and $d\mathcal{J}/dt$ derived above.

- (c.4): Augment the state vector by the work state ($\mathbf{Q}_D$: drag-derived generalized force)

$$\dot{\mathbf{W}}_D = \mathbf{Q}_D^\top \dot{\mathbf{q}}, \quad \mathbf{W}_D(0) = 0 \quad (7)$$

and **propagate** the orbit under drag again. **Plot** the normalized work-balance residual

$$\frac{\mathcal{J}(t) - \mathcal{J}(0) - \mathbf{W}_D(t)}{|\mathcal{J}(0)|}. \quad (8)$$

Discuss whether it is consistent with your integrator tolerance.

- (d): **Optional (extra credit), [AI use Level: 1]: Newtonian approach in spherical coordinates.**

- (d.1): **Derive** the same spherical-coordinate EOMs from Newton's second law.
- (d.2): **Show** that the result is identical to the EOM derived using Lagrange's equations.
- (d.3): **Discuss** advantages of each of the two approaches: Lagrangian and Newtonian approaches.

Problem Set 6

(6.1) Attitude Dynamics with Solar Arrays and Reaction Wheels

Consider a satellite consisting of a rigid bus B , two identical rigid solar arrays A_1 and A_2 , and three reaction wheels, in a torque-free environment. The system center of mass C is fixed in the inertial frame $\mathcal{N}$. A body-fixed frame $\mathcal{B} : \{\hat{\mathbf{b}}_1, \hat{\mathbf{b}}_2, \hat{\mathbf{b}}_3\}$ is attached to the bus at C . The configuration is detailed below. Fig. 1 shows a schematic illustration of the spacecraft.

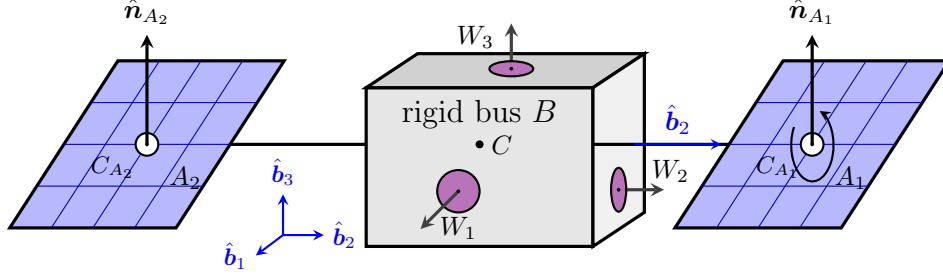


Figure 1: Schematic illustration of the spacecraft considered in PS6.

Solar arrays: The array centers, C_{A_1} and C_{A_2} , are fixed relative to the bus at

$$\mathbf{r}_{C_{A_1}/C} = d\hat{\mathbf{b}}_2, \quad \mathbf{r}_{C_{A_2}/C} = -d\hat{\mathbf{b}}_2. \quad (9)$$

Each array rotates about a boom axis parallel to $\hat{\mathbf{b}}_2$ and passing through its own center of mass. Since the solar panels need to keep tracking the Sun, both panels have the same normal vector direction:

$$\hat{\mathbf{n}}_{A_1} = \hat{\mathbf{n}}_{A_2} = \mathbf{R}_2(\beta)\hat{\mathbf{b}}_3, \quad (10)$$

where $\mathbf{R}_2(\cdot)$ is a matrix representing a rotation about $\hat{\mathbf{b}}_2$. Assume the panel rotation angle β is prescribed:

$$\beta(t) = \begin{cases} \beta_m (10\tau^3 - 15\tau^4 + 6\tau^5), & 0 \leq t \leq t_m, \\ \beta_m, & t > t_m, \end{cases} \quad \tau := \frac{t}{t_m}. \quad (11)$$

where t_m, β_m are the time and angle at the end of the maneuver, respectively, and τ is normalized time.

Denote by $\mathbf{I}_{A0}$ the inertia of each panel about its own center of mass when $\beta = 0$. Thus, the panel inertia with the rotation β is:

$$\mathbf{I}_A(\beta) = \mathbf{R}_2(\beta)\mathbf{I}_{A0}\mathbf{R}_2^\top(\beta). \quad (12)$$

Reaction wheels (RWs): Wheel W_i has a fixed spin axis $\hat{\mathbf{b}}_i$, axial moment of inertia J_{w_i} , and spin angle α_i , for $i = 1, 2, 3$. Thus, relevant terms are defined as:

$$\boldsymbol{\alpha} = [\alpha_1, \alpha_2, \alpha_3]^\top, \quad \boldsymbol{\Omega} := \dot{\boldsymbol{\alpha}}, \quad \mathbf{J}_w := \text{diag}(J_{w_1}, J_{w_2}, J_{w_3}). \quad (13)$$

Spacecraft bus: Denote the bus inertia about C by $\mathbf{I}_B$. It includes the inertia of the booms and the wheel transverse inertia components.

Express the orientation of $\mathcal{B}$ relative to $\mathcal{N}$ using (3-1-3) Euler angle sequence, defined as $\boldsymbol{\theta} = [\theta_1, \theta_2, \theta_3]^\top$. The relation between the time derivative of $\boldsymbol{\theta}$ and the angular velocity of $\mathcal{B}$ relative to $\mathcal{N}$ is given by:

$$\boldsymbol{\omega} := {}^{\mathcal{B}}\boldsymbol{\omega}_{B/N} = \mathbf{E}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}}, \quad \mathbf{E}(\boldsymbol{\theta}) = \begin{bmatrix} \sin \theta_3 \sin \theta_2 & \cos \theta_3 & 0 \\ \sin \theta_2 \cos \theta_3 & -\sin \theta_3 & 0 \\ \cos \theta_2 & 0 & 1 \end{bmatrix}. \quad (14)$$

Generalized coordinates: Express the generalized coordinates in terms of the bus orientation and wheel spin angles:

$$\mathbf{q} = [\boldsymbol{\theta}^\top, \boldsymbol{\alpha}^\top]^\top. \quad (15)$$

(a): [AI use Level: 2] Lagrange model.

(a.1): **Discuss** why it makes sense to not include $\beta(t)$ in the generalized coordinates Eq. (15).

- (a.2): Let us derive kinetic energy for each component, i.e., bus, solar array assembly, and RW assembly. Denoting them by T_B, T_A , and T_W , respectively, **show** that they are given by:

$$T_B = \frac{1}{2} \boldsymbol{\omega}^\top \mathbf{I}_B \boldsymbol{\omega}, \quad T_A = \frac{1}{2} (\boldsymbol{\omega} + \dot{\beta} \hat{\mathbf{b}}_2)^\top \mathbf{I}_{A/C}(\beta) (\boldsymbol{\omega} + \dot{\beta} \hat{\mathbf{b}}_2), \quad T_W = \frac{1}{2} (\boldsymbol{\omega} + \boldsymbol{\Omega})^\top \mathbf{J}_w (\boldsymbol{\omega} + \boldsymbol{\Omega}) \quad (16)$$

where $\mathbf{I}_{A/C}(\beta)$ is the combined array inertia about C , given by:

$$\mathbf{I}_{A/C}(\beta) = 2\mathbf{I}_A(\beta) + 2m_A d^2 (\mathbf{I}_3 - \hat{\mathbf{b}}_2 \hat{\mathbf{b}}_2^\top) \quad (17)$$

Hint: use the parallel axis theorem to derive $\mathbf{I}_{A/C}$.

- (a.3): Noting that the total kinetic energy is $T = T_B + T_A + T_W$, **derive** the Lagrangian L for this system and **express** it in a form that explicitly depends on $(\mathbf{q}, \dot{\mathbf{q}})$ by eliminating $\boldsymbol{\omega}, \boldsymbol{\Omega}$ from L .
- (a.4): Let u_i be the motor torque applied to W_i , and $\mathbf{u} = [u_1, u_2, u_3]^\top$. **Derive** the EOM using Lagrange's equations, and **show** the result in a vector-matrix form: $\mathbf{M}(\mathbf{q}, t) \ddot{\mathbf{q}} = \mathbf{f}(\mathbf{q}, \dot{\mathbf{q}}, t, \mathbf{u})$.

Hint 1: the following terms help simplify the derivation.

- Total spacecraft angular momentum about C : $\mathbf{H}_S = \mathbf{I}_B \boldsymbol{\omega} + \mathbf{I}_{A/C}(\boldsymbol{\omega} + \dot{\beta} \hat{\mathbf{b}}_2) + \mathbf{J}_w (\boldsymbol{\omega} + \boldsymbol{\Omega})$;
- Instantaneous total spacecraft inertia about C : $\mathbf{I}_S = \mathbf{I}_B + \mathbf{I}_{A/C} + \mathbf{J}_w$.

Hint 2: notice that $\mathbf{I}_{A/C}$ is a function of time, but you can leave $\dot{\mathbf{I}}_{A/C}$ in the solution for brevity.

- (a.5): **Show** that $\boldsymbol{\alpha} \in \mathbb{R}^3$ is cyclic, and **derive** its generalized momentum, denoted by $\mathbf{p}_\alpha \in \mathbb{R}^3$.
- (a.6): **Derive** the time derivative of $\mathbf{p}_\alpha$, and **discuss** when it is conserved.
- (a.7): **Calculate** the Jacobi integral $\mathcal{J}$ and its time derivative, and **discuss** if it is conserved.

(b): **[AI use Level: 2] Numerical investigation: Zero RW torque.**

Consider the attitude behavior with zero RW motor torque $\mathbf{u}(t) = \mathbf{0}$ with the initial spin rate $\boldsymbol{\Omega}(0) = \boldsymbol{\Omega}_0$, while the solar arrays are rotating according to Eq. (11). You can use Eq. (18) without derivation:

$$\dot{\mathbf{I}}_{A/C}(t) = 2\dot{\mathbf{I}}_A(t) = 2\dot{\beta}(t) \left([\hat{\mathbf{b}}_2]_\times \mathbf{I}_A(t) - \mathbf{I}_A(t) [\hat{\mathbf{b}}_2]_\times \right). \quad (18)$$

For the numerical work, use the following parameters and initial conditions:

$$\begin{aligned} \mathbf{I}_B &= \text{diag}(80, 100, 60) \text{ kg m}^2, \quad m_A = 24 \text{ kg}, \quad d = 1.5 \text{ m}, \\ \mathbf{I}_{A0} &= \text{diag}(6, 30, 35) \text{ kg m}^2, \quad \mathbf{J}_w = \text{diag}(0.10, 0.12, 0.08) \text{ kg m}^2, \\ \beta_m &= 60 \text{ deg}, \quad t_m = 60 \text{ s}, \\ \boldsymbol{\theta}(0) &= [\pi/3, \pi/3, \pi/3]^\top \text{ rad}, \quad \boldsymbol{\omega}(0) = \mathbf{0}, \quad \boldsymbol{\alpha}(0) = \mathbf{0}, \quad \boldsymbol{\Omega}(0) = \boldsymbol{\Omega}_0 = [300, -200, 100]^\top \text{ rad/s} \end{aligned} \quad (19)$$

- (b.1): **Plot** the time history of β and $\dot{\beta}$ based on Eq. (11).
- (b.2): **Implement** the EOM derived above, and **report** the implemented code. Numerically **propagate** the system for $0 \leq t \leq 100$ sec with $\mathbf{u} = \mathbf{0}$. **Plot** the time histories of $\boldsymbol{\theta}, \boldsymbol{\omega}$, and $\boldsymbol{\Omega}$.
- (b.3): **Plot** the time history of $\mathbf{p}_\alpha$, and **discuss** its time variation and why the result makes sense. Also, **discuss** why $\boldsymbol{\Omega}(t)$ is generally not constant.
- (b.4): **Optional (extra credit):**
From the symbolic expression of $\mathbf{p}_\alpha$, **derive** the relation between $\boldsymbol{\Omega}(t)$ and $\boldsymbol{\omega}(t)$ under $\mathbf{u}(t) = \mathbf{0}$. **Verify** that this analytical relation between $\boldsymbol{\Omega}(t)$ and $\boldsymbol{\omega}(t)$ is consistent with the numerical result.

(c): **[AI use Level: 1] Numerical investigation: Arbitrary RW torque.**

Let us then simulate the spacecraft attitude for an arbitrary RW torque of your choice.

- (c.1): First, **define** and **report** the RW torque as a smooth function of time $\mathbf{u}(t)$, and **plot** its time history over $0 \leq t \leq 60$ sec.

(c.2): Numerically **propagate** the system for the same time span using the EOM implemented above. **Plot** the time histories of $\boldsymbol{\theta}$, $\boldsymbol{\omega}$, $\boldsymbol{\Omega}$, and $\mathbf{p}_\alpha$. **Discuss** the result and why it makes sense.

(d): **Optional (extra credit), [AI use Level: 2]: Newtonian (Eulerian) approach.**

Discuss a high-level sketch of the derivation procedure for the EOMs of this system using the Euler's rotational equations of motion, **show** the complete derivation, and **verify** its consistency with the EOM derived above.